

Support Social Connectedness through Technology-mediated Shared Activities

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ABSTRACT

Shared activities can improve the quality of connection during mediated communication and help overcome barriers with traditional communication technology. Prior research has explored new technology designs to support shared activities and context in remote communication. Only a few studies have investigated the long-term impact of novel technology design on people's social connectedness. My dissertation work explores technology design that support in-person social practices in remote context during shared activities, and studies the long-term effect of the proposed system design on people's social connections. The four different studies are designed to understand current practices with shared activities in building and maintaining social connections, design communication technology to nurture meaningful relationship during remote activities and evaluate the proposed system with a long-term field study. Overall, my dissertation contributes an in-depth understanding of how technology can better support people's social practices during shared activities and helps inform opportunities for technology to better connect people.

KEYWORDS

social connectedness, intergenerational communication, shared activities, remote play, collaborative reading

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1 INTRODUCTION

Strong social connections and inter-personal support mitigate the risk of social isolation and improve people's health and wellbeing [3]. Shared activities play an important role in people's in-person social practices: playing boardgames with friends, attending happy hours with colleagues, or reading books with family. Such activities provide a common ground for communication and can often improve social outcomes from communication. Shared activities and context are especially important for intergenerational communication, since children often face a variety of challenges when

using traditional communication technologies [1]. Researchers have designed new technology systems to support shared context and activities for better communication quality (e.g., one that supports sharing location-based stories among family members [2]). However, most of the prior work focuses on short-term evaluations with their proposed technology design and lacks an understanding of the long-term effect on people's social connectedness. My dissertation work explores the current in-person practices with shared activities that connect people socially, translate the in-person practices to remote technology design, and evaluate the impact of proposed design on people's social connectedness through a long-term field study. Specifically, my dissertation studies aim to address the following research questions:

(RQ1) What are people's needs and priorities involved in current intergenerational communication practices?

(RQ2) What social technical factors affected people's remote playing and social experiences?

(RQ3) How might technology better support people's social connectedness during remote shared activities?

2 RESEARCH STUDIES OVERVIEW

In this section, I provide an overview of four studies from my dissertation work. For each study, I describe the context of the investigation and the study methods. I also summarize the findings of the studies that have been completed.

2.1 Contextual Inquiry with Older Adults and Children (RQ1)

Volunteering programs like intergenerational mentorship provide opportunities for older adults to connect with the younger generation at a community level and strengthen both generations' bonds with their local communities. Such programs often connect older adults and children through collaboration on coursework (e.g., reading, writing and math tutoring). However, current infrastructures with intergenerational mentorship programs place the burdens of logistics and transportation on older adults as they often require in-person visits. To understand how technology can help expand access to intergenerational mentoring and translate in-person activities to online format, we partnered with a local non-profit organization and conducted a contextual inquiry study of their ongoing intergenerational mentorship program. We observed eight intergenerational mentoring sessions with four older adult volunteers and 10 children in three local elementary schools. We also conducted in-depth interviews with the four volunteers and one teacher to understand their social connectedness expectations. Our data revealed three design opportunities for technology to support intergenerational connection through mentoring activities:

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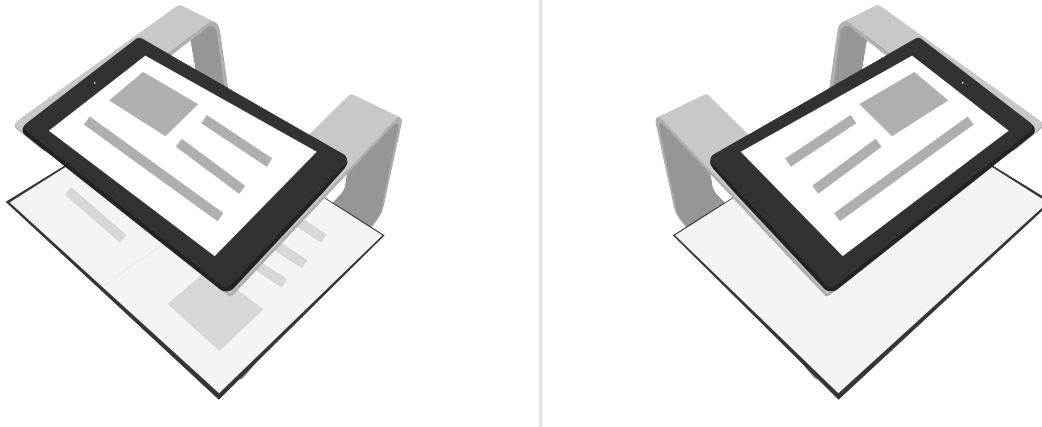


Figure 1: Tablet-based media space includes a tablet running the application and a physical customized stand, creating a shared space below the display. Physical objects (e.g., physical book on the left of the figure) and gestures placed in the shared space will be combined and presented on both displays.

1) lowering barriers for adults to participate in the activities, 2) nurturing meaningful relationship during shared activities, and 3) increasing facilitator's capacity for organizing intergenerational mentoring activities.

2.2 Remote Play Observation and Interview Study (RQ2)

Tabletop gaming (e.g., boardgames, card games) is a common social practice among family and friends. However, during the recent coronavirus disease 2019 (COVID-19) pandemic, in-person social activities were restricted because of the social distancing rules in countries across the world. To maintain connections with family and friends, people have managed to use available resources to move their play sessions from offline to online. This study aims to understand the factors that affect people's social experience during remote play when transitioning from offline to online tabletop gaming. We used a qualitative approach in this study, combining a three-month online observation and 15 in-depth interviews. We found three themes that characterized people's social and game experience in remote play setup: 1) creating a shared tabletop space for remote play, 2) building a collective understanding between players through communication and awareness, and 3) facilitating a communal temporal experience during remote play. Our findings highlight the importance of physical and tangible components in remote tabletop gaming, as people often use a hybrid approach to construct their shared tabletop space (e.g., sharing physical game components through phone cameras). These insights suggest opportunities in technology to support sharing tangible objects and natural interactions when designing systems to support shared activities that rely on physical objects in offline settings (e.g., reading and playing).

2.3 Technology Design and Prototyping (RQ3)

Based on the design opportunities identified in section 2.1 and 2.2 for technology to better support social connectedness through shared

activities, we designed a tablet-based system that allows sharing physical objects and hand gestures. The design of the system is inspired by projector-based media space (e.g., [4]). The proposed system (Figure 1) consists of a web application and a physical stand. The physical stand constructs a space for users to share objects and gesture interactions. The tablet display serves as an augmented lens of the shared space below the tablet. The augmented view of the shared space supported by the app is prototyped using WebRTC and image processing technologies. One of the intended usage scenarios with this system is remote reading, which is a common practice in intergenerational communication. The following section provides more information about how the proposed technology design will be deployed and evaluated in the field, in the context of supporting social connectedness through remote reading activities between older adults and children.

2.4 Field Deployment Study (RQ3)

To understand how the proposed technology design in section 2.3 might support people's social connectedness through remote shared activities, we plan to run a field deployment study with three older adults and six children. We will have them use the system prototype to complete eight remote reading sessions (each session lasts about 15-20 minutes) during two months, since reading is identified as one of the important activities that provides shared context for intergenerational communication and foster social connectedness in section 2.1. We will conduct semi-structured interviews at the beginning of the study, to train participants on the technology usage, ask questions about their relationship and expectations with the technology. We will also meet with our participants weekly, to ensure that all participants can use the system and complete the activities. At the end of the study, we will have in-depth interviews with participants to solicit feedback on the technology. Given the findings from section 2.1, we will also recruit two teachers as research partners and experts in activity content, who will help provide feedback on the study design, identify potential child

participants, and facilitate the study sessions. We plan to conduct two workshops with teachers to collaborate on the study design and planning before the recruitment starts. For this field deployment study, the majority of the data collected from the field will be qualitative from interviews, while some quantitative data will be collected from the system log. This study will help us understand RQ3, with rich usage data from deploying the proposed system in the field.

3 WORK IN PROGRESS

By the time this abstract is written, the studies outlined in section 2.1 and section 2.2 are completed and published in 2019 [6] and 2021 [5] at CHI. We have been working with the system prototype described in section 2.3 and preparing the prototype for the field deployment study (section 2.4). We have connected with local elementary school teachers and the non-profit organization who runs the intergenerational mentorship program to work together on the study planning and participants recruiting. We plan to start recruiting participants in August 2021 and complete the field study before March 2022. We plan to complete the analysis of the study in Summer 2022.

4 EXPECTED CONTRIBUTIONS

My dissertation studies intend to make the following contributions: 1) an empirical understanding of people's practices with shared activities and factors that affect people's social experiences in these shared activities; 2) a technology design and proof-of-concept prototype to support people's social connectedness through shared

activities; and 3) an understanding of how technology design affects people's social experience during technology-mediated shared activities. In general, my dissertation helps better understand the impact of technology-mediated communication on people's social connectedness and informs future social technology design.

REFERENCES

- [1] Rafael Ballagas, Joseph 'Jofish' Kaye, Morgan Ames, Janet Go, and Hayes Raffle. 2009. Family communication: phone conversations with children. In *Proceedings of the 8th International Conference on Interaction Design and Children (IDC '09)*. Association for Computing Machinery, New York, NY, USA, 321–324. <https://doi.org/10.1145/1551788.1551874>
- [2] Frank R. Bentley and Sujoy Kumar Chowdhury. 2010. Serendipitous family stories: using findings from a study on family communication to share family history. In *Proceedings of the 12th ACM international conference adjunct papers on Ubiquitous computing - Adjunct (UbiComp '10 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 359–360. <https://doi.org/10.1145/1864431.1864435>
- [3] J. S. House, K. R. Landis, and D. Umberson. 1988. Social relationships and health. *Science* 241, 4865 (July 1988), 540–545. <https://doi.org/10.1126/science.3399889> Publisher: American Association for the Advancement of Science Section: Articles.
- [4] Svetlana Yarosh, Anthony Tang, Sanika Mokashi, and Gregory D. Abowd. 2013. "almost touching": parent-child remote communication using the sharetale system. In *Proceedings of the 2013 conference on Computer supported cooperative work (CSCW '13)*. Association for Computing Machinery, New York, NY, USA, 181–192. <https://doi.org/10.1145/2441776.2441798>
- [5] Ye Yuan, Jan Cao, Ruotong Wang, and Svetlana Yarosh. 2021. Tabletop Games in the Age of Remote Collaboration: Design Opportunities for a Socially Connected Game Experience. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Number 436. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3411764.3445512>
- [6] Ye Yuan and Svetlana Yarosh. 2019. Beyond Tutoring: Opportunities for Intergenerational Mentorship at a Community Level. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300679>